

Approximation algorithms with predictions

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Selection problems

Items $i \in [n]$ have nonnegative weights $w(i)$.

An instance specifies a family $\mathcal{F}$ of feasible subsets of $[n]$.

Choose $X \in \mathcal{F}$ to minimize or maximize $w(X) = \sum_{i \in X} w(i)$.

The family $\mathcal{F}$ can be exponentially large. Feasibility depends on the instance, while the objective adds item weights.

The input is available offline, so the paper calls the quality measure an approximation ratio.

EXAMPLES

Four selection problems

Minimum vertex cover

Items: vertices
Feasible: every edge has an endpoint selected
Minimize selected vertex weight

Minimum perfect matching

Items: edges
Feasible: each vertex meets exactly one selected edge
Minimize selected edge weight

Maximum clique

Items: vertices
Feasible: every two selected vertices are adjacent
Maximize selected vertex weight

Knapsack

Items: objects
Feasible: total size is at most capacity
Maximize selected object value

Prediction and error

The predictor supplies a set P of items believed to occur in an optimal solution O .

P may be infeasible. The algorithm must return a feasible set.

$$\eta^+ = w(P \setminus O)$$

false positives: predicted items outside O

$$\eta^- = w(O \setminus P)$$

false negatives: optimal items omitted by P

$OPT = w(O) > 0$ for the ratios below.

THEOREM 1

Minimization: discount predicted items

Start with any ρ -approximation algorithm A for the original selection problem, where $\rho \geq 1$.

Modified weight

$$\bar{w}(i) \quad 0 \quad \text{if } i \in P$$

$$w(i) \quad \text{if } i \notin P$$

Run A on the same feasible family with modified weights $\bar{w}$. Return its feasible approximate solution X .

THEOREM 1

Minimization guarantee

For every feasible comparison set X' :

$$w(X) \leq w(X') + \eta^+(X') + (\rho - 1)\eta^-(X')$$

Choosing an optimal O gives an error-sensitive approximation bound:

$$\frac{w(X)}{OPT} \leq 1 + \frac{[\eta^+ + (\rho - 1)\eta^-]}{OPT}$$

Here $\eta^+ = w(P \setminus O)$ and $\eta^- = w(O \setminus P)$.

Where the two errors enter

A sees only items outside P as costly:

$$w(X \setminus P) = \bar{w}(X) \leq \rho \bar{w}(O) = \rho w(O \setminus P) = \rho \eta^-$$

Restore the original cost of items selected from P :

$$w(X) = w(X \cap P) + w(X \setminus P) \leq w(P) + \rho \eta^-$$

Since $w(P) = OPT + \eta^+ - \eta^-$, we obtain $OPT + \eta^+ + (\rho - 1)\eta^-$.

Consistency and robustness

Perfect prediction

$$P = 0 \Rightarrow \eta^+ = \eta^- = 0 \Rightarrow w(X) = OPT$$

Worst prediction

η^+ may be arbitrarily large, so this version has no constant cap.

Offline fallback

Run A also on w ; return the cheaper solution: $\text{ratio} \leq \min\{\rho, 1 + [\eta^+ + (\rho - 1)\eta^-]/OPT\}$.

Application: minimum vertex cover

Choose vertices covering every edge of an undirected graph.

A standard weighted vertex-cover algorithm gives $\rho = 2$.

Predict $P \subseteq V$, set its vertex weights to zero, and run the 2-approximation.

$$\frac{w(X)}{OPT} \leq 1 + \frac{\eta^+ + \eta^-}{OPT}$$

Running the ordinary 2-approximation as well caps the ratio at 2.

A partial discount: $\lambda \in (0, 1]$

Modified weight

$\bar{w}(i)$

$$\lambda w(i) \quad \text{if } i \in P$$

$$w(i) \quad \text{if } i \notin P$$

$$\frac{w(X)}{OPT} \leq \min \left\{ \frac{\rho}{\lambda}, 1 + (\rho - 1)\lambda + (1 - \lambda) \frac{[\eta^+ + (\rho - 1)\eta^-]}{OPT} \right\}$$

Why λ is unnecessary for the generic offline guarantee

Write $B_0 = 1 + [\eta^+ + (\rho - 1)\eta^-]/\text{OPT}$ for the paper's zero-discount bound.

$$B\lambda = (1 - \lambda)B_0 + \lambda\rho$$

The λ error bound lies between B_0 and the baseline ratio ρ .

Zero discount + original-weight fallback: ratio $\leq \min\{B_0, \rho\}$

This guarantee is at least as strong as λ 's $\min\{B\lambda, \rho/\lambda\}$.

THEOREM 3

Maximization via the complementary problem

A feasible maximization set X corresponds to its complement $Y = [n] \setminus X$.

The complementary problem is:

minimize $w(Y)$ subject to $[n] \setminus Y$ being feasible for the original problem

Assume a ρ -approximation A for minimum-weight feasible complements Y .

Modified weight $\bar{w}(i) = w(i)$ if $i \in P$

0 if $i \notin P$

Run A to find Y using $\bar{w}$; return $X = [n] \setminus Y$.

Complement feasibility ensures X is feasible, even when P is not.

THEOREM 3

Maximization guarantee and proof idea

$$w(X) \geq OPT - (\rho - 1)\eta^+ - \eta^-$$

The complement of O is feasible, so the baseline removes at most $\rho\eta^+$ of predicted weight.

The prediction itself carries $w(P) = OPT + \eta^+ - \eta^-$.

$$\frac{w(X)}{OPT} \geq 1 - \frac{[(\rho - 1)\eta^+ + \eta^-]}{OPT}$$

If this lower bound is negative, nonnegative weights still imply a fraction of at least 0.

Maximization examples

Maximum clique

In the complement graph, a clique becomes an independent set. Its omitted vertices form a vertex cover. A 2-approximate minimum vertex cover gives fraction $\geq 1 - (\eta^+ + \eta^-)/\text{OPT}$.

Knapsack

The omitted items must have total size at least $\sum s(i) - \text{capacity}$. Approximate the minimum value discarded, then keep the remaining items. The paper gives a 2-approximation for this complementary problem.

With an ordinary maximization baseline of fraction ρ' , choosing the better solution caps the loss: fraction $\geq \max\{\rho', 1 - [(\rho - 1)\eta^+ + \eta^-]/\text{OPT}\}$.